

Keil μ Vision Templates



F. Angione, P. Bernardi, R. Ferrero

Politecnico di Torino

Dipartimento di Automatica e Informatica (DAUIN)

Torino - Italy

This work is licensed under the Creative Commons (CC BY-SA) License. To view a copy of this license, visit <http://creativecommons.org/licenses/by-sa/3.0/>



What is needed

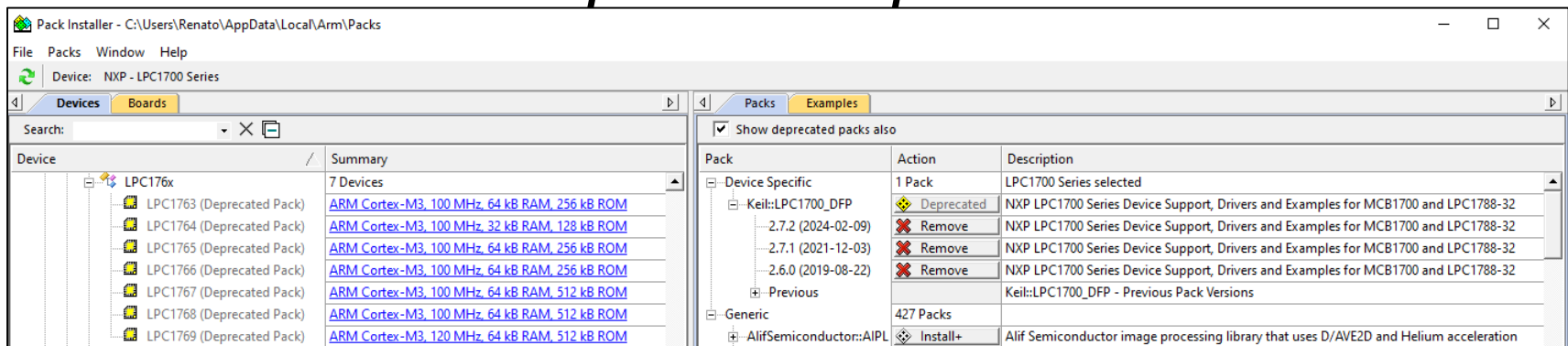
- Keil μ vision v5.43: clean installation suggested
- ARMCLANG v6.24: it comes with Keil v5.43.
- ARM::CMSIS v6.2.0
 - installed by Keil Pack Installer when loading a project.
- LPC1700 Device Family Pack (DFP) v2.7.2
- Landtiger Emulator

LPC1768 device pack

- The device pack for the LPC1700 family, which includes LPC1768, is available here:

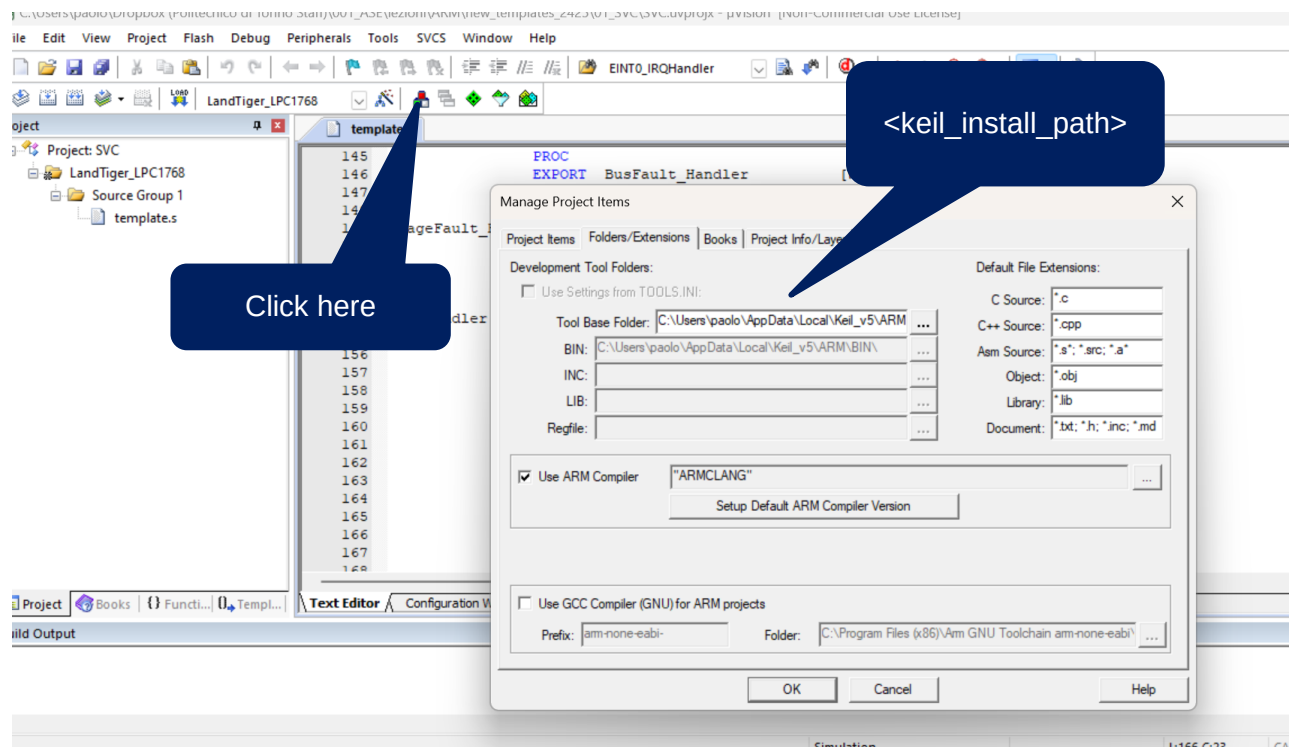
https://www.keil.arm.com/packs/lpc1700_dfp-keil/versions/

- To install the package, open Pack Installer, go to *File -> Import...* and then select the location where you saved the device pack
- To see the package in the left windows, check *Show deprecated packs also*



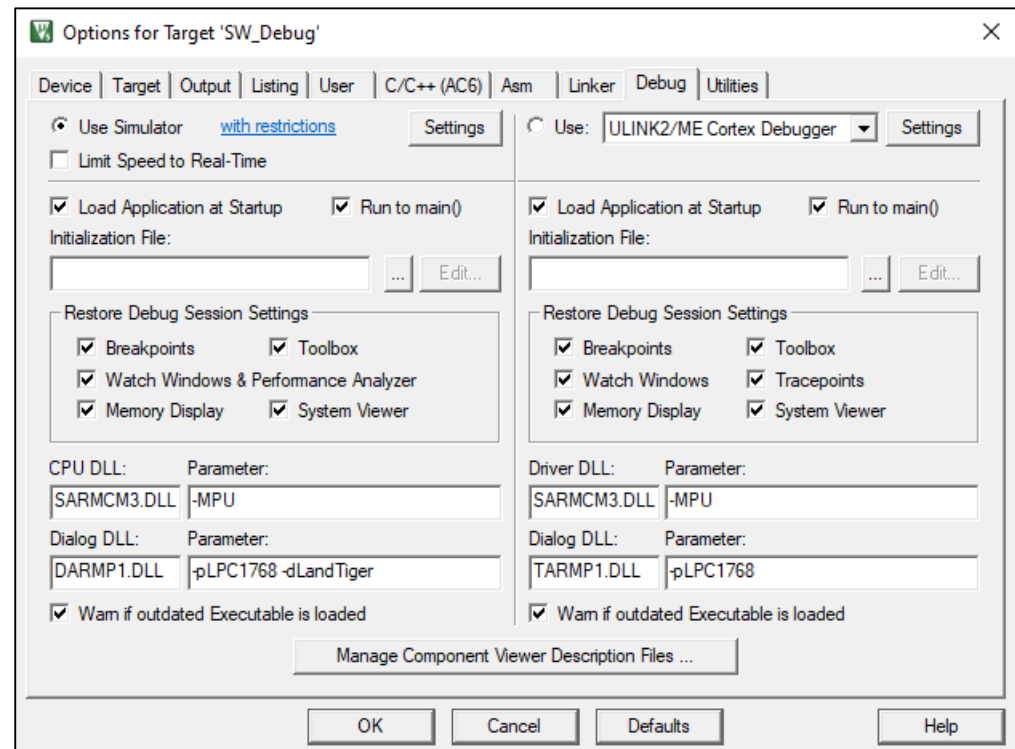
Landtiger Emulator

- Preliminary step: install VC_redist.x86.exe
- Place the files extracted from LandTiger.zip in <KEIL_INSTALL_PATH>_v5\ARM\BIN



How to use the Landtiger Emulator

- In *Options for Target*, go to the *Debug* tab
- On the *Parameter* textbox next to *Dialog DLL* type *-dLandTiger*
 - only for the left section *Use simulator*
- The parameter is already set in the target *SW_Debug* provided with the templates.



How to use the μ Vision Simulator

- A built-in instruction set simulator is available for some cores (including Arm Cortex M3)

<https://developer.arm.com/documentation/101407/0543/Simulation/uVision-Simulation-Targets?lang=en>

- To simulate the behavior of on-chip peripherals, you need to specify the Device Simulation in *Options for target -> Debug*

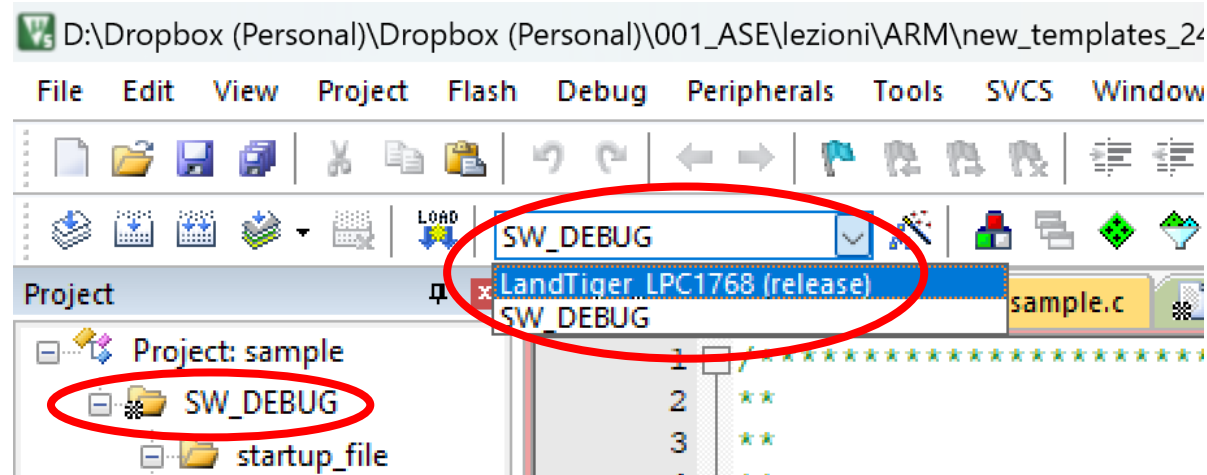
- Dialog DLL: *DARMP1.DLL* (Simulator) or *TARMP1.DLL* (ULINK2/ME Cortex Debugger)
- Parameter: *-pLPC1768* (both)

<https://developer.arm.com/documentation/ka002225/latest>

- The parameters are already set in the templates

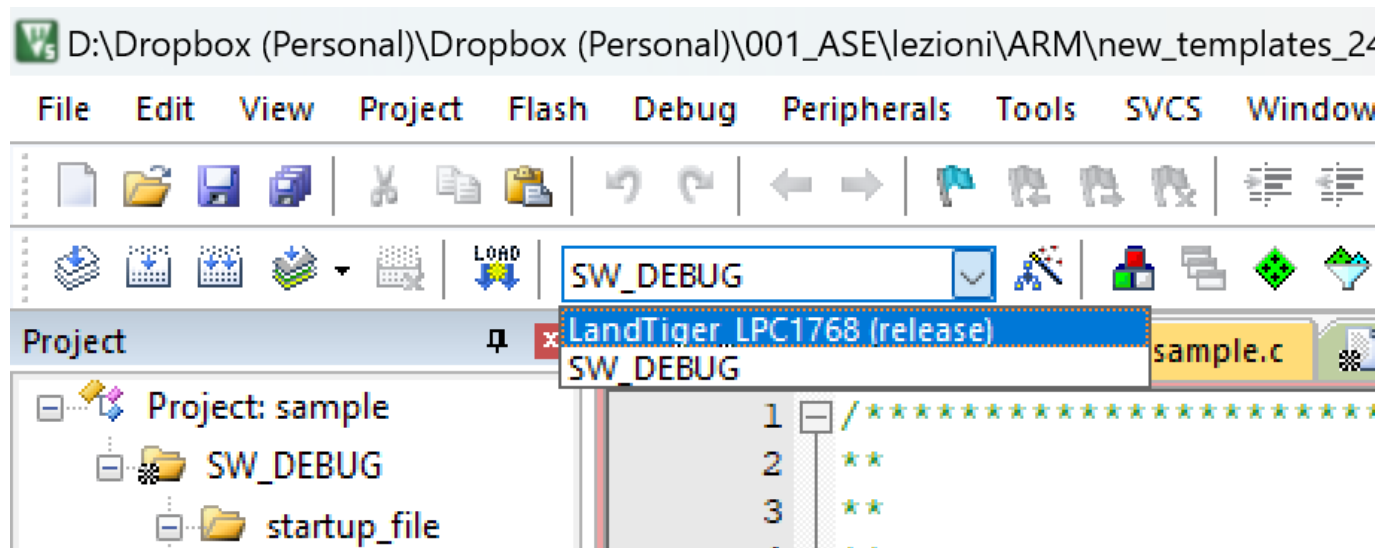
Project Targets

- When a new project is created there a default target named *Target1*
- The templates substitute it with new targets:
 - LandTiger_LPC1768 (release)
 - SW_DEBUG
- You can add and modify new target with *Manage project items...*



Different targets for different goals

- Select the target depending on your needs:
 - SW_DEBUG: run the program using the software simulator/emulator.
 - LandTiger_LPC1768 (release): run the program on the real board



Target tab

Options for Target 'LandTiger_LPC1768'

Device **Target** Output Listing User C/C++ (AC6) Asm Linker Debug Utilities

NXP LPC1768

Xtal (MHz): <undefined>

Operating system: None

System Viewer File: LPC176x5x.svd

☐ Use Custom File

Code Generation

ARM Compiler: V6.22

☐ Use MicroLIB ☐ Big Endian

Read/Only Memory Areas

	default	off-chip	Start	Size	Startup
☐		ROM1:			☐
☐		ROM2:			☐
☐		ROM3:			☐
		on-chip			
☒		IROM1:	0x0	0x80000	☒
☐		IROM2:			☐

Read/Write Memory Areas

	default	off-chip	Start	Size	NoInit
☐		RAM1:			☐
☐		RAM2:			☐
☐		RAM3:			☐
		on-chip			
☒		IRAM1:	0x10000000	0x8000	☐
☒		IRAM2:	0x2007C000	0x8000	☐

OK Cancel Defaults Help

Compiler tab: Includes

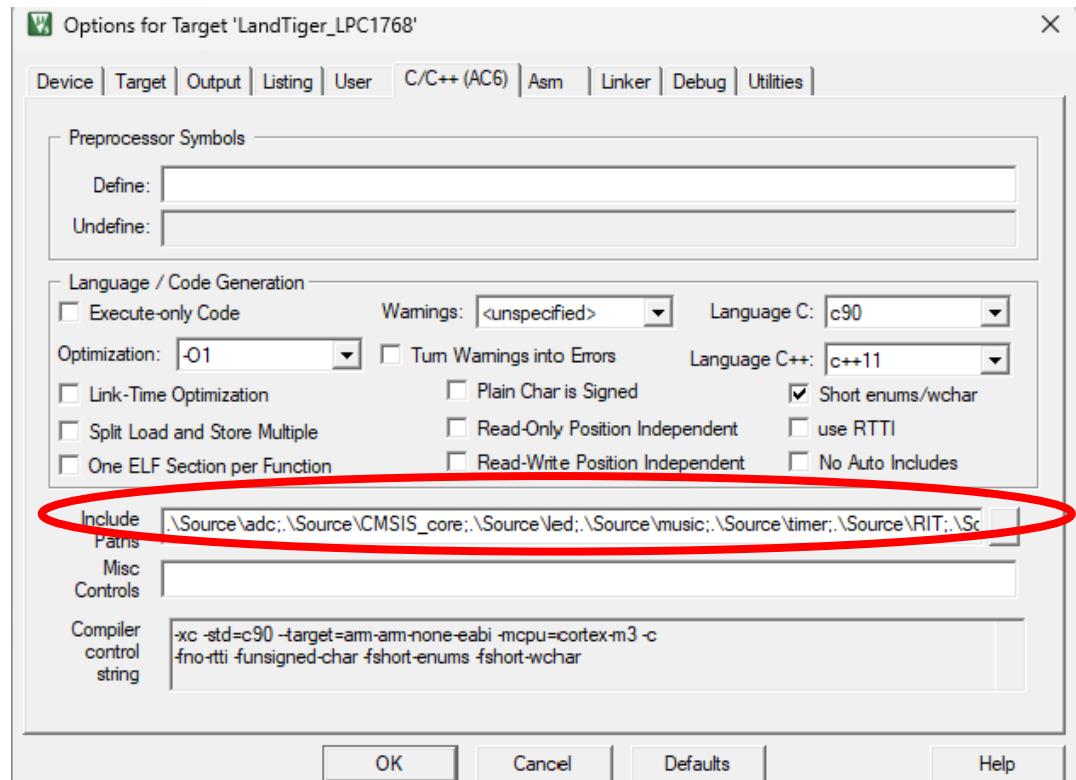
- New folders must be added in this list:

- *Source*
- subdirectories of *Source*

- The folders are separated with ;

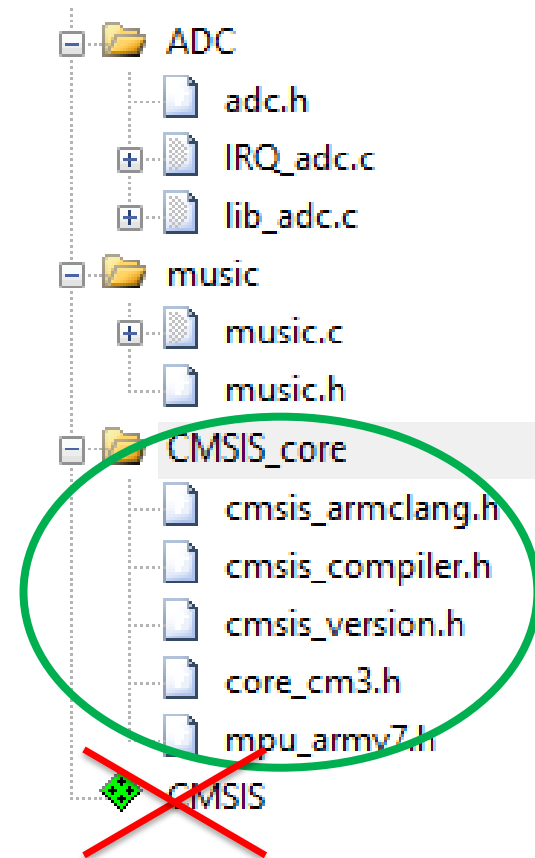
- Example

.\Source;
.\Source\led;
.\Source\button_EXINT;
.\Source\CMSIS_core



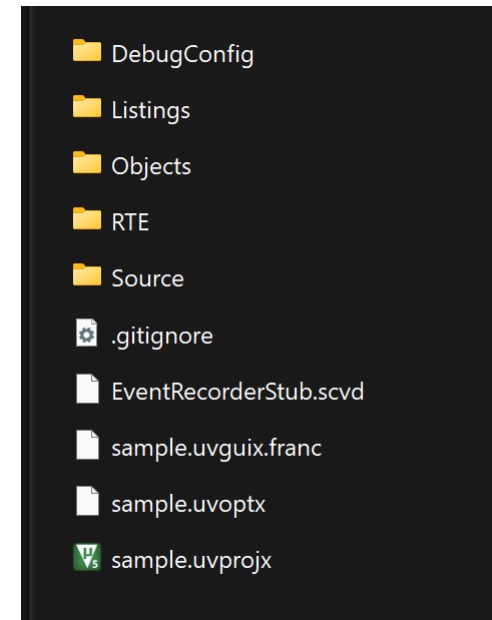
The Common Microcontroller Software Interface Standard (CMSIS)

- The project files include *CMSIS_core*
- It is an alternative to  *CMSIS* included from RTE (Runtime Environment)
 - You can remove it in Project -> Manage -> Run-Time environment, by deselecting CMSIS -> CORE
- Dependencies in DFP_LPC1700 are included for fixing



The project structure

- Compilation directories:
 - *Objects*: *.o, *.d, *.map, *.axf and other compilation outputs
 - *Listings*: *.dep, *.lst
- Debug directories:
DebugConfig and *.scvd
- Runtime Environment: *RTE*
- Source files in *Source*
- Keil projects: *.uvoptx, *.uvproj



Reuse GUI configuration

- The templates come with the GUI configuration of another user (different from you): e.g. *sample.uvguix.franc*
- If you don't rename it, a new file will be created for each user.

